ACTIVE SPEAKER – ESP32-WROVER / PCM5122 / TPA3126D2

12–16 V DC -> protected PVDD -> stereo BTL amplifier.
PVDD -> 4.30 V FPWM buck -> ESP32 3.3 V LDO + DAC 3.3 V low-noise LDO.
PCM5122 LEFT = woofer / CH1; RIGHT = tweeter / CH2. Both speaker returns are active.
External Nichicon UES capacitor on J18; replace with shorting link for stereo use.
No passive crossover. DSP crossover/EQ/limiting remains a firmware and acoustic task.

POWER

PVDD
4V_PRE
3V3-ESP
VIN_SENSE

H1 MountingHole
H2 MountingHole
H3 MountingHole
H4 MountingHole

File: Power.kicad_sch

ESP32

3V3-ESP
VIN_SENSE
BCKD
LRCKD
DATAD
SCLD
SDA
DAC_XSMT
AMP_ENABLE
AMP_MUTE_CMD
AMP_FAULT_N

File: ESP32.kicad_sch

DAC

4V_PRE
BCK
LRCK
DATA
SCL
SDA
DAC_XSMT
WOOFER_AUDIO
TWEETER_AUDIO

File: DAC.kicad_sch

AMP

PVDD
3V3-ESP
WOOFER_AUDIO
TWEETER_AUDIO
AMP_ENABLE
AMP_MUTE_CMD
AMP_FAULT_N

File: Amp.kicad_sch

STARTUP: rails settle -> ESP32 boots (SDZ low, MUTE high, XSMT low).
Configure PCM5122 at 0x4C with both channels muted; configure DSP and valid 64fs I2S.
Enable SDZ, wait >=50 ms, release DAC mute, then amplifier mute; ramp DAC volume.
SHUTDOWN: ramp down, assert DAC XSMT low >=150/fs +0.2 ms; assert MUTE then SDZ low.
FAULT: mute and shut down, log fault; deliberate retry only after fault cause is removed.
Review report and bench validation are required before connecting the real drivers.

HS1 Heatsink

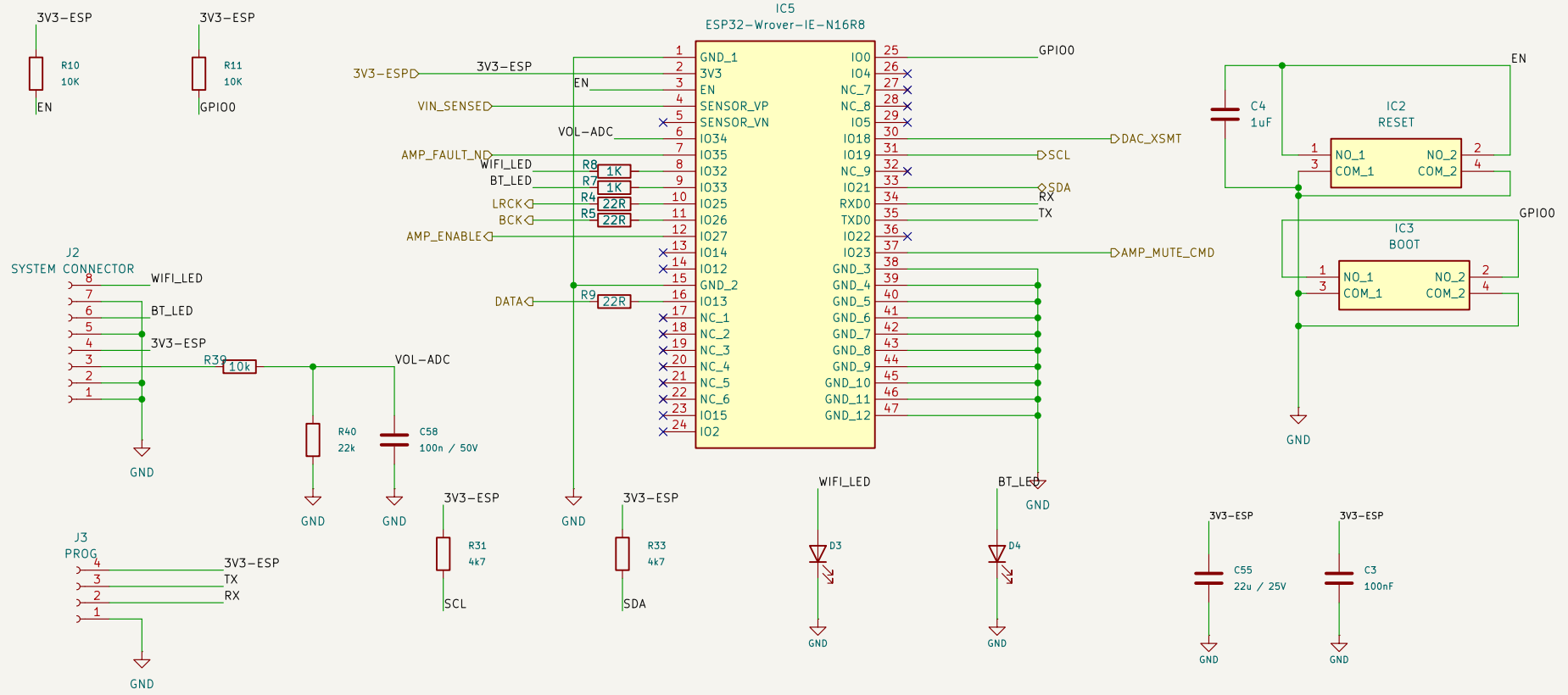
Schematic completion only – PCB unchanged

Sheet: /
File: TPA3126.kicad_sch

Title: Bluetooth / Snapcast active speaker

Size: A3	Date: 2026-10-19	**Rev: Schematic review A**
KiCad E.D.A. 10.0.5		Id: 1/5

ESP32-WROVER-IE-N16R8: EXTERNAL ANTENNA



Preserved: LED Wi-Fi=32, BT=33; LRCK=25, BCK=26, DIN=13; SCL=19, SDA=21.
Added: SDZ enable=27; MUTE command=23 (HIGH=mute); FAULTZ=35 (input only); DAC_XSMT=18.
GPIO36 ADC1 monitors PVDD/11. Volume pot: GPIO34 / ADC1_CH6 (module pin 6). GPIO0 BOOT pull-up; EN reset has 10k/1u RC.
GPIO6-11 / GPIO16-17 reserved for internal flash/PSRAM: no external use.
J2 system: 2=GND, 3=volume wiper, 4=3V3. Suggested pot: 10k linear.
J3 is the separate programming header; use 3.3 V UART logic.
Module custom pads 39-47 are segments of the common exposed GND area.
Fit a compatible external antenna to the IE module; keep RF connector/feed away from switch nodes.

VOLUME POT: J2 pins 2=GND, 3=wiper, 4=3V3. Use 10k linear.
R39/R40 scale full travel to ~2.27V; C58 filters ADC noise.
GPIO34 = ADC1_CH6. Firmware: highest ADC attenuation, average readings;
calibrate endpoints / taper and map low or disconnected wiper to mute.

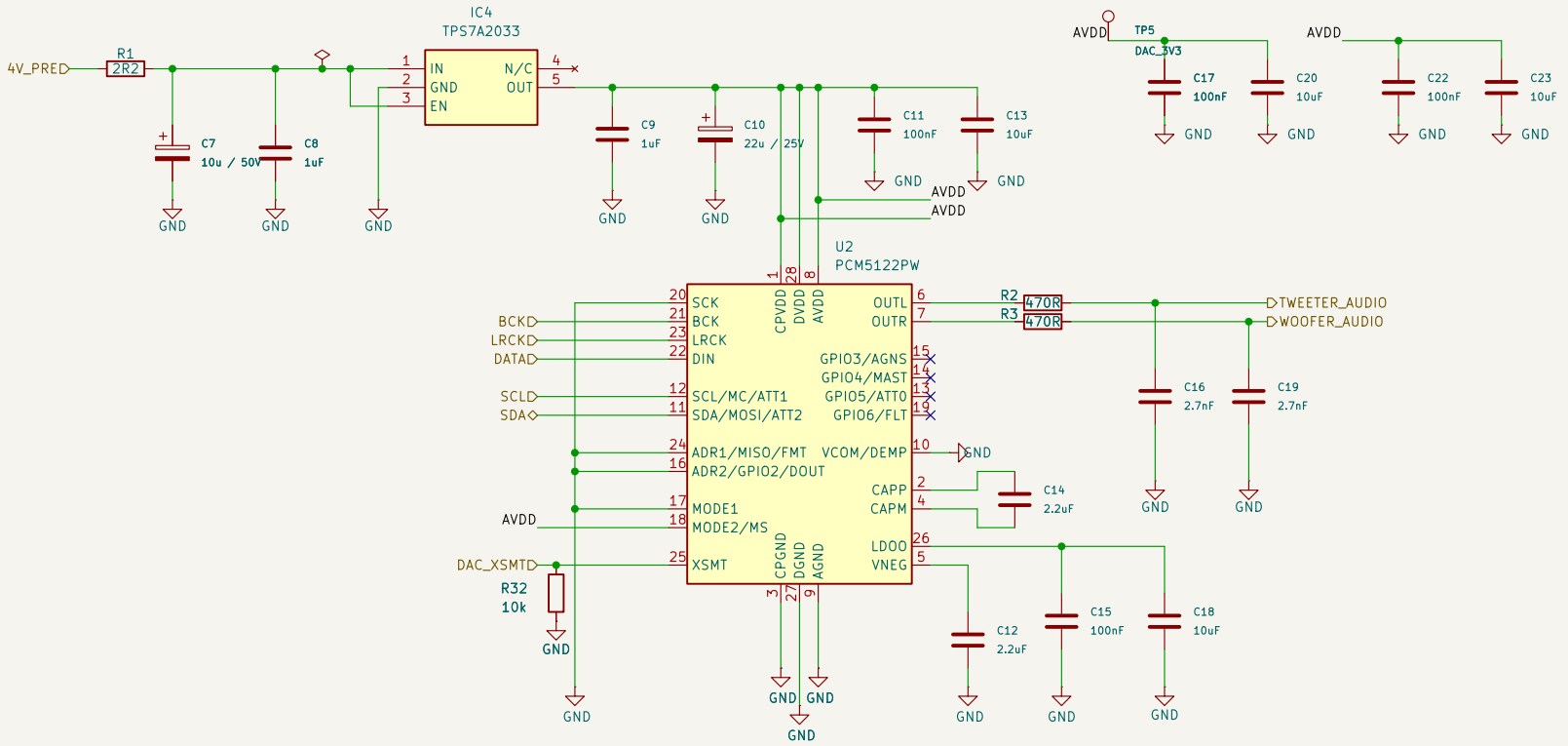
Schematic completion only - PCB unchanged

Sheet: /ESP32/
File: ESP32.kicad_sch

Title: ESP32-WROVER-IE - retained firmware GPIOs

Size: A4	Date: 2026-10-19	**Rev: Schematic review A**
KiCad E.D.A. 10.0.5		Id: 2/5

PCM5122 – I2C VOLUME / MCLK-LESS I2S



MODE1=0, MODE2=1; ADR2=ADR1=0 -> I2C 0x4C (7 bit). SCK grounded.
BCK -> PLL reference; use 64fs BCK and I2S 32-bit slots. Enable normal clock-error handling.
P0 R13 SRCREF=BCK (0x10). P0 R61/R62 = channel digital volume; ramp together for master volume.
OUTL -> woofer (TPA right bridge); OUTR -> tweeter (TPA left bridge).
R2/R3 + C16/C19 retain 470R/2n7 C0G output filtering; amp input pads limit full-scale level.
XSMT pulled low by R32 until GPIO18 releases mute. Keep volume muted throughout setup.
VCOM grounded in default VREF mode. LDOO decoupled; unused GPIO3-6 are no-connects.
Clock loss automatically powers down DAC/line drivers. Valid but incorrect DSP data still needs limiting.
All grounds share GND; route analogue return to source, away from amplifier switching currents.

Schematic completion only – PCB unchanged

Sheet: /DAC/
File: DAC.kicad_sch

Title: PCM5122 – left woofer, right tweeter, register controlled

Size: A4	Date: 2026-10-19	**Rev: Schematic review A**
KiCad E.D.A. 10.0.5		Id: 3/5

TPA3126D2 – STEREO BTL / 4 OHM LOADS

20 dB MASTER; AM[2:0]=000 -> 400 kHz nominal. MODSEL=0 -> hybrid mode.
Each CSAD package: winding 1–2, winding 4–3. No assumed flux cancellation.
TI Fig.31 filter: 10u per leg + 680n to GND per leg (four film capacitors total).
Keep 20 x 18 x 10 mm heatsink clearance over U1; top PowerPAD, no PCB EP copper.
Top pad may be grounded (preferred) or floating per TI; never connect it to PVDD.

PVDD
C57 1nF
C56 1nF
C35 220uF
C33 100nF
C36 220uF
C34 100nF
GND

U1
TPA3126D2DADR

PVCC 18
PVCC 19
PVCC 31
PVCC 32
AVCC 17
GVDD 7
RINP 4
RINN 5
LINP 10
LINN 11
MODSEL 1
SDZ 2
PLIMIT 6
GAIN_SLV 8
MUTE 12
AM2 13
AM1 14
AM0 15
SYNC 16

OUTPR 29
OUTNR 27
BSNR 26
BSPR 30
BSPL 24
BSNL 20
OUTPL 23
OUTNL 21
FAULTZ 3
GND 9
GND 22
GND 25
GND 28

C28 1uF
AMP_RINP
AMP_RINN
AMP_LINP
AMP_LINN
AMP_SDZ
PLIMIT
GAIN_SLV
AMP_MUTE

SW_WP
SW_WN
SW_TP
SW_TN
C25 220nF
C24 220nF
C26 220nF
C27 220nF
AMP_FAULT_N

L2 10uH
L3 10uH
TP9 W+
TP10 W-
TP11 T+
TP12 T-
C37 680nF
C38 680nF
C39 680nF
C40 680nF
TW_FILTER_P
TW_PROTECTED_P
J16 WOOFER / CH1
J17 TWEETER / CH2

JP1 Jumper
C21 47uF

SAFE STARTUP / FAULT

3V3-ESP
AMP_ENABLED
R21 10K
TP7 SDZ
R25 100K
C30 100n / 50V
R29 100K
TP8 MUTE
R26 10K
C31 100nF
R30 100k
TP6 FAULTZ

SDZ default LOW; MUTE default HIGH. 10k/100n command filtering
limits slew below TI 10 V/ms. SDZ asserted low is the primary safe state.
MUTE command HIGH=mute, LOW=play. Wait >=50 ms after SDZ high
before unmuting. FAULTZ open-drain -> GPIO35, 100k pull-up.
No automatic retry link: log fault, mute, cycle SDZ only after diagnosis.

C41 330p DNP
C42 330p DNP
C43 330p DNP
C44 330p DNP
R35 18R DNP
R36 18R DNP
R37 18R DNP
R38 18R DNP

EXTERNAL TWEETER CAPACITOR / STEREO BYPASS

C54 = Nichicon UES1H470MPM, 47u / 50V bipolar, 10 x 20 mm.
Wire across J18 pins 1–2. It is in series with CH2 positive.
For stereo: remove capacitor harness and fit a rated shorting link.
With neither cap nor link, CH2 positive is OPEN (safe no-output).
Pin 1 = filtered bridge positive; pin 2 = speaker connector positive.
Neither pin is ground. Change link only with power off.
Nominal 4 ohm pole ~847 Hz; actual response follows tweeter impedance.
Account for capacitor phase/response in the DSP crossover.

15 W / CHANNEL TARGET AT 16 V

PLIMIT: 3V3-ESP -> 10k / 1u RC; R20 DNP.
TI peak-rail model ~19W unclipped / 4 ohm nominal.
Headroom for 15W; not an exact 15W protection limit.
Input pad 1k2 / 1k5 plus DAC 470R -> 0.466 gain.
15W / 4 ohm = 7.75Vrms: calibrate DSP per channel.
Power derates below 16V; measure clipping / thermals.

R19 10K
R20 33k DNP
C29 1uF
R18 5K6
R13 1K5
R12 1K2
C1 1uF
C2 1uF
R16 820R
WOOFER_AUDIO
AMP_RINP
AMP_RINN

SNUBBERS C41–C44 / R35–R38: DNP, tune on bench.

TI starting point 330p / 18R; no copied TPA3220 snubber.
All filtered speaker terminals float relative to GND.
Route source-side input reference returns to the DAC GND.
Power loops must not share those return traces.

R15 1K5
R14 1K2
C6 1uF
C3 1uF
R17 820R
TWEETER_AUDIO
AMP_LINP
AMP_LINN

WOOFER: DAC LEFT -> AMPLIFIER RIGHT BRIDGE

TWEETER: DAC RIGHT -> AMPLIFIER LEFT BRIDGE

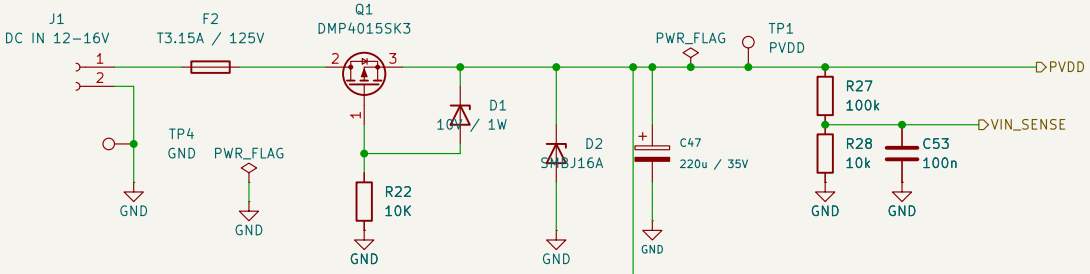
Schematic completion only – PCB unchanged

Sheet: /AMP/
File: Amp.kicad_sch

Title: TPA3126D2 – stereo BTL, bypassable CH2 capacitor

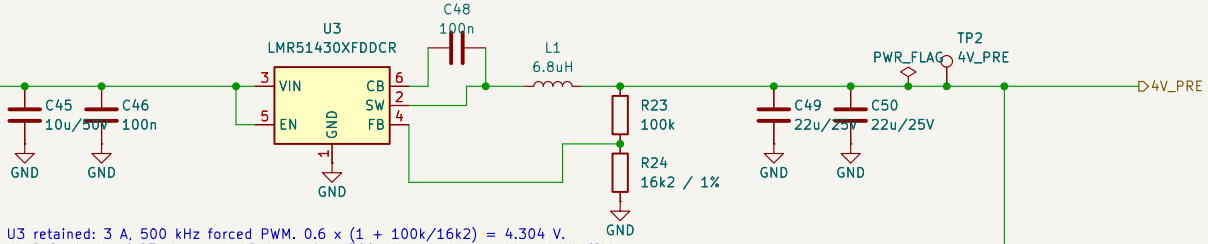
Size: A3 | Date: 2026-10-19 | **Rev: Schematic review A**
KiCad E.D.A. 10.0.5 | Id: 4/5

INPUT: 12–16 V DC ONLY



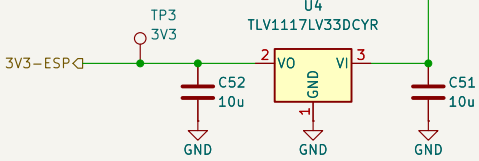
J1 pin 1 = +DC; pin 2 = GND. F2: T3.15A. Q1 reverse polarity; D1 clamps |VGS|.
D2 SMBJ16A: 16 V standoff, 26 V rated pulse clamp. Not a sustained overvoltage regulator.
C47 input bulk; C35/C36 are local amplifier bulk. Use a regulated, current-limited SELV supply.
R27/R28: PVDD/11 -> GPIO36; 100n filtering. ADC1 works while Wi-Fi is active.

4.30 V PRE-REGULATOR



U3 retained: 3 A, 500 kHz forced PWM. 0.6 x (1 + 100k/16k2) = 4.304 V.
L1 6u8: ripple ~0.93 A p-p at 16 V, nominal 500 kHz. Local ceramics at VIN/SW.
Feeds both 3.3 V LDOs. Do not substitute the PFM LMR51430X version.
VIN_SENSE is advisory: mute/shut down early during a falling input; verify hold-up on bench.

3.3 V ESP32 LDO



TLV1117LV33: 1 A; tab = output. Retained topology, reduced input voltage.
At 0.25 A: ~0.25 W; at 0.5 A: ~0.50 W; at 1 A: ~1.00 W.
Use copper heat spreading. C52 10u ceramic plus C55 22u at ESP32.
TPS7A2033 on DAC sheet supplies only the DAC.

Schematic completion only – PCB unchanged

Sheet: /POWER/
File: Power.kicad_sch

Title: Power – protected input, 4.30 V, 3.3 V

Size: A4	Date: 2026-10-19	**Rev: Schematic review A**
KiCad E.D.A. 10.0.5		Id: 5/5